UMHackathon 2026: Preliminary Round Deliverable

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Project Name: SmartStock AI

Group Members

Name	Role	University
Maha Vaarshinie A/P Arupatha Rajoo	Team Leader	Multimedia University
Nareenthiran A/L Bhuvaneswaran	Team Member	Multimedia University
Yasha Durgashine	Team Member	Multimedia University

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1. Project Overview

- **Problem Statement:**

  The difficulties of the Small and Medium Enterprises (SMEs), such as local cafes, convenience stores, and retail shops, with their issue of inventory management and cost control, are common. Most of these businesses are based on manual estimation, intuition, or some simple spreadsheet tools that are not predictive. This means that they usually face the challenge of over-stocking, hence spoilage of products and unnecessary holding costs, or under-stocking, hence loss of sales opportunities and customer dissatisfaction.

  Moreover, SMEs usually lack access to complex analytical tools that are able to include external influencing variables like weather conditions, seasonal trends or local events. Such contextual variables have a huge influence on customer demand and are usually ignored because of the non-integrated, clever systems. This leaves the decision-making process reactive instead of proactive, constraining the business in terms of optimizing its operations and maximizing its profitability.

- **Target Domain:**

  The current project works in the framework of AI-based decision intelligence systems to improve the work of SMEs. In particular, it aims at optimizing inventory and cost management, using artificial intelligence to turn raw information into actionable information. It is intended to facilitate the need to close the gap between the availability of data and the ability to make decisions to allow SMEs to leverage both structured data (e.g., sales records, inventory levels) and unstructured data (e.g., textual notes, external contextual factors) in a single analytical framework. Through this, it enables non-technical users to make decisions that are based on the data without the need for sophisticated analytical skills.

- **Proposed Solution Summary:**

  SmartStock is an artificial intelligence-based decision support system, created to help the SME owners streamline inventory management processes and minimize operational

expenses. The system combines the various sources of data, such as structured data, such as CSV files with sales and inventory data of the past, and unstructured data, such as user-entered notes about external factors (e.g., heavy rain tomorrow or local festival next weekend).

The Z is at the heart of SmartStock.The core reasoning engine is known as AI General Language Model (GLM). The model converts both structured and unstructured data to detect patterns, contextual information, and produce demand predictions. According to this analysis, the system gives recommendations in actionable form, which could be a change in order quantities, lowering stock levels of perishable goods, or getting ready against a rise in demand following a given event.

Besides producing recommendations, the system also gives explicit, human-understandable explanations to its decisions, which make the system transparent and create trust in the user. Moving inventory planning away from a manual, reactive process to an intelligent, proactive process, SmartStock allows SMEs to reduce waste and stock shortages, not to mention better operational efficiency. The system has quantifiable effects in terms of reduction of inventory expenses, reduction in product wastage, and saving of time in making decisions.

2. Background & Business Objective

- **Background of the Problem:**

  The methods used in inventory planning in Small and Medium Enterprises (SMEs) are usually traditional, experience-driven, as opposed to systematic, data-driven methods. Business owners usually base their decisions on intuition, historical patterns, or just spreadsheet tracking, which makes them incapable of making accurate predictions about demand or adapting to evolving circumstances. SMEs, in most instances, lack access to sophisticated analytical applications or technical skills to read sales trends, seasonality, and other external-influencing factors like weather, holidays, or local events. Consequently, decision-making becomes reactive as opposed to proactive. This results in typical inefficiencies in operational practices, such as spoiled products as a result of over-stocking, stock outages resulting in lost revenue, and needless man-hours in manual planning.

  These problems are further enhanced in fast-moving industries like food and beverage or retail, where demand can vary greatly within short periods. In the absence of knowledgeable support systems, SMEs find it hard to achieve equilibrium between supply and demand, which eventually impacts their profitability as well as survival.

- **Importance of Solving This Issue:**

  Addressing inefficiencies in inventory management is critical for improving the financial and operational performance of SMEs. By transitioning to a more intelligent and
  With a data-driven approach, businesses can achieve measurable benefits, including:

  1. **Reduction in operational costs** through minimized waste, particularly for perishable goods.
  2. **Increased revenue** by ensuring optimal stock availability and avoiding missed sales opportunities.
  3. **Improved productivity and time efficiency** by automating manual planning and reducing reliance on guesswork.

Beyond immediate financial gains, improving decision-making capabilities also enhances business resilience, enabling SMEs to better adapt to market changes and external disruptions.

- **Strategic Fit / Impact:**
  SmartStock directly aligns with the core objective of UMHackathon 2026, which emphasizes AI-powered decision intelligence for economic empowerment. The solution empowers SMEs by transforming raw data into actionable insights, allowing them to make informed decisions that were previously inaccessible without advanced tools or expertise.

  The system delivers impact in the following ways:

  - Enabling smarter, data-driven decision-making for non-technical users
  - Reducing financial losses caused by inefficient inventory practices
  - Improving operational efficiency through AI-generated insights and recommendations
  - Democratizing access to AI technology, allowing small businesses to benefit from capabilities typically available only to larger enterprises

  By focusing on a highly practical and widespread problem, SmartStock has the potential to create meaningful, real-world economic impact at scale.

3. Product Purpose

3.1. Main Goal of the System:

The main aim of SmartStock is to offer SMEs an AI-powered decision intelligence system to optimize inventory planning and cost. The system will be developed in such a manner that it will be able to use past data and real-time contextual inputs to come up with the correct demand forecasts and suggest the best inventory decisions. Instead of substituting the role of human decision-making, SmartStock serves as a smart assistant that supplements the judgment of the

user by displaying information-based insights and scenario analysis and presenting a clear explanation. This will help to keep users in control whilst enjoying the power of AI.

3.2. Intended Users (Target Audience):

SmartStock is targeted at a wide group of SME stakeholders, especially those who do not have access to advanced analytical tools or technical knowledge.

The target users are:

- Small business owners (e.g., cafe owners, owners of retail shops) who make direct inventory decisions and manage costs.

The SME managers are in charge of daily operations and in need of fast and dependable information to sustain efficiency. Inventory planners and personnel have only basic technical skills and require easy-to-use and intuitively designed tools to aid in decision-making. The system is user-friendly and accessible, so that a user can communicate with the platform using the simplest (e.g., CSV uploads or natural language notes) inputs without the need to be trained in the system.

4. System Functionalities

4.1 Description

SmartStock is an artificial intelligence-based decision intelligence software that can be used by SMEs to make optimal inventory decisions. The system incorporates both structured data (e.g., sales records, stock levels) and unstructured inputs (e.g., user notes about weather, events, or future trends) into a single analytical pipeline.

Essentially, the system takes advantage of the Z.AI General Language Model (GLM) to carry out contextual reasoning and predictive analysis. SmartStock does not just show historical data but converts raw data into proactive information, such as demand predictions, inventory-optimization plans, and cost-cutting ideas.

The system focuses on explainability, which means that each recommendation should have a clear explanation. This will enable the users to know the logic behind AI-generated decisions and trust the system. In its absence, the system would only be able to display the data in a static manner, thus not helping in making any meaningful decisions.

4.2 Key Functionalities

1. Data Entry (CSV Upload / Manual Entry)

This module enables the user to add both quantitative and contextual information:

- Post organised data (CSV format) with:
- Historical sales data
- Inventory levels
- Product categories

Enter informational content out of context, like:

- Weather (e.g., heavy rain, heatwave, etc.)
- Local events (e.g., festivals, promotions)
- Business-related notes (e.g., anticipated customer rush).

This two-input methodology will make sure the system is able to capture both the numerical trends and the real world.

2. Predictive Trends Dashboard

This part acts as the data visualization layer, acting as a central hub for monitoring the performance of the business:

- **7-Day Demand vs Sales Visualizer:** Uses Recharts for visualizing "AI Predicted Demand" vs. "Historical Sales" in order to offer immediate evidence for the forecast model's accuracy.
- **Spoilage Risk Index:** Dynamic bar graph that evaluates the probability of spoilage of inventory due to current stocks versus projected number of foot traffic.
- **User-Friendly Interface:** A straightforward and easy-to-navigate interface meant for non-tech savvy SME owners for identifying both low- and high-demand products.

3. AI Model Tuning and Guardrails

In order for the SME owner to feel comfortable and in control, there will be a model tuning section for:

- **AI Risk Tolerance Slider:** Gives the user freedom in determining the aggressiveness of the AI's inventory trimming (i.e., from conservative/safe to aggressive/growth).
- **"What If" Scenario Simulator:** A traffic drop simulator which helps users manually stress test the assumptions made by the AI prior to creating decision matrix.

4. AI Insights & Recommendations Panel.

This is the essence of the decision-supporting part, which is driven by the GLM:

Demand Prediction: Predicts short-term demand using past trends and contextual information.

Inventory Recommendations: Recommends the best stock quantities and reorder quantities.

Cost Optimization Strategies: Establishes waste reduction opportunities and holds cost reduction opportunities.

Explainability Layer: Will give clear, human-readable explanations of each recommendation.

Example output: Cut down the number of milk orders by half tomorrow because of the likely low demand due to heavy rainfall.

4.3 AI Model & Prompt Design

4.3.1 Model Selection

Model: Z.Artificial intelligence (AI) General Language Model (GLM)

Justification:

- Developed thinking skills in decision-making.
- Capability to work with both structured (table) and unstructured (text) data.
- Produces explainable outputs and is context-aware.
- Mandatory requirement according to hackathon requirements.

The GLM is closely knit into the reasoning layer as the main computational element, and it cannot be done without the functionality of the system.

4.3.2 Prompting Strategy

SmartStock uses a multi-step (agentic) prompting approach to enhance the accuracy, reliability, and interpretability of AI-generated outputs. Rather than passing the raw user input to the Z. The system, AI GLM, works with the data in a structured pipeline, with each step of processing the data having a particular purpose of preparation and refinement of the input before it produces the final recommendation. This graded methodology assists in making sure that the model will generate context-sensitive and logically coherent outputs.

The prompting process has three key steps:

I. Data Summarization Stage

During this step, unstructured data (e.g., CSV files with sales and inventory records) is converted into a succinct and meaningful summary. Major trends, including average sales, high demand times, and product trends, are derived. This minimizes data complexity that is unnecessary, and only the relevant information is forwarded to the model, maximizing the use of tokens and enhancing processing time.

II. Context Interpretation Stage

The system then interprets unstructured data that the user feeds it, including notes on weather patterns or events ahead. These inputs are understood and translated into action. An example is when someone states that there will be heavy rain tomorrow; this is translated into an implied effect of low customer foot traffic. This is done to enable the model to reflect real-life factors that are not directly described in structured data.

III. Decision Generation Stage

Finally, the structured information and context interpretation are merged to form a properly structured prompt and then passed to the GLM. To ensure that the stability of the system is not compromised during frontend rendering, the Z.AI GLM is programmed using the system prompt to generate a properly validated JSON object. It is clearly directed by the system prompt to organize its reasoning in certain keys:

- **mock_strategy:** The key strategy and quantities for stock pivoting.
- **mock_tradeoff:** The rationale behind the risks of the recommended pivot.
- **riskLevel:** An integer value between 0-100 representing the spoilage risk vector.

4.3.3 Context & Input Handling

In order to be scaled, efficient, and meet model constraints, SmartStock adopts a systematic process of managing both structured and unstructured inputs. As large inputs may adversely affect performance and surpass the token limits, the system preprocesses the data to optimize it and then transmits it to the GLM.

The system works within the input constraints as follows:

- **Maximum Input Limits:**
  - CSV data: 5,000 rows (MVP limit) or more.
  - Text input: Around 1,000-2,000 tokens.

  The system uses various strategies to deal with the inputs that are above these limits:

- **Data Chunking**

  Bulky data is broken down into smaller and more manageable parts. The segments are then acted upon individually and then amalgamated into an encompassing representation. This makes sure that the system is capable of processing voluminous data without putting a strain on the model.

- **Data Summarization**

  The historical information is summarized into important insights (e.g., averages, trends, and anomalies) to minimize the usage of tokens. This makes sure that only the most pertinent information is contained in the prompt.

- **Input Filtering & Truncation**

  Too long or unrelated user inputs are filtered or cut off to eliminate noise. This allows ensuring a timely clarity and makes sure that the model dwells upon the significant context only. On the whole, this controlled input manipulating system makes the system

efficient and responsive, and within token constraints. It also enhances the quality of AI outputs because it feeds the model with clean, relevant, and well-structured input data.

4.3.4 Fallback & Failure Behavior

I. **To ensure reliability and user trust, SmartStock has powerful error-handling features:**
   - Checking of invalid or hallucinated responses with structured response validation (JSON schema).
   - Refined prompts and automatic retry mechanism.

II. **Graceful fallback response, e.g.:**
   - Can no longer form sound insight. Refine your input.

III. **User control:** Gives the user the capability to re-enter or modify inputs.

IV. This will guarantee the predictable behavior of the system despite uncertain AI outputs.

5. User Stories & Use Cases

5.1 User Stories

User stories tell us about the system as seen by the end users and the purpose and value they expect to see in the system. In SmartStock, the user stories are tailored to demonstrate the actual needs of the SME owners, who need to have simple yet powerful tools to make their decisions.

1. **User Story 1**

"I am a cafe owner and would like to post my sales data in order to be given correct inventory recommendations".

This user story indicates the necessity of using data to make decisions. Several owners of SMEs already possess access to sales records, but they do not have the tools to analyze these records. Allowing users to add structured data (in the form of CSV files) helps SmartStock recognize

patterns and trends and respond to changes in demand. This is the basis of creating correct inventory suggestions, assisting users in decreasing wastage and preventing shortages.

2. **User Story 2**

"I would like to add some contexts (e.g., weather, events) as a shop owner so that the predictions were made in real-life conditions".

The relevance of context-aware intelligence is highlighted in this user story. External forces tend to affect business demand and cannot be reflected solely in historical data. SmartStock can use these aspects of real-world variables by letting users add unstructured notes, like an upcoming event or the weather condition, to its predictions. It makes sure that recommendations are not solely premised on historical data but also in tandem with the present and future circumstances.

3. **User Story 3**

"As a user, I would like to understand AI recommendations straightforwardly to trust and authenticate the decisions made by the system".

This user story is concerned with the transparency and trust in AI systems. When they comprehend the manner in which such decisions are made, users will be more inclined to embrace and trust AI-generated recommendations. SmartStock solves this by giving clear definitions that are easy to understand and read with every suggestion. This will enable users to authenticate the logic, to gain confidence in the system, and make informed decisions instead of blindly following AI results.

5.2 Use Cases (Main Interactions)

1. Data Upload Flow

User uploads CSV data → System validates and stores data → Backend prepares input → GLM processes data → Insights are generated and displayed

2. Insight Generation Flow

User enters contextual notes → System combines structured and unstructured data → GLM performs reasoning → Generates predictions and recommendations → Outputs displayed with explanation.

3. Dashboard Interaction

User accesses dashboard → Views stock levels and trends → Reviews AI-generated insights → Takes action based on recommendations

5.3 Use Case Diagram

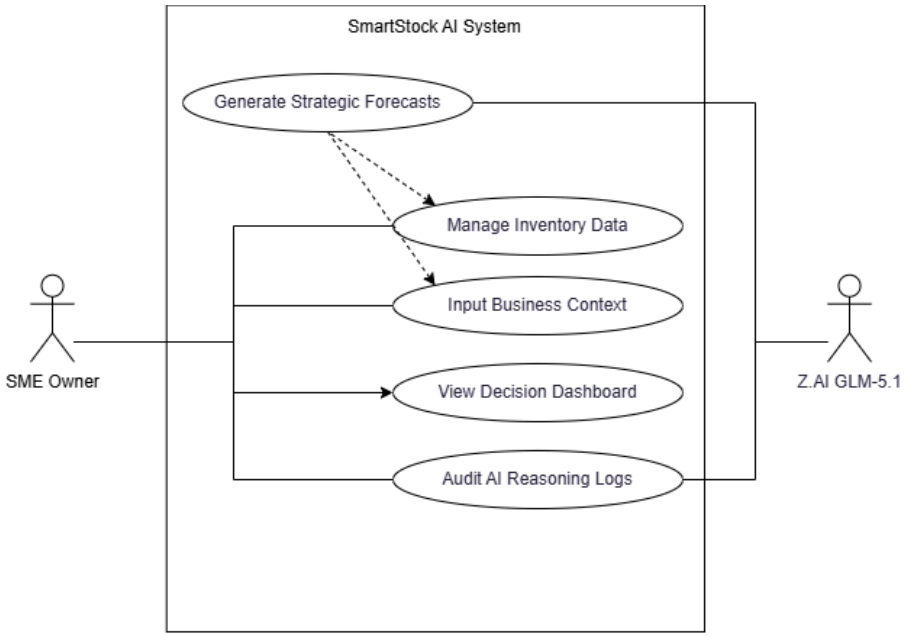

Figure 1: Use Case Diagram

The SmartStock AI Use Case Diagram is utilized as a working blueprint that describes the limits of the system and the mutually reinforcing relationship between the human user and the intelligent layer that is essential. The SME Owner serves as the main participant, who launches the necessary data-driven objectives, like maintaining inventory records, adding unformatted contextual notes, and observing business trends, through a centralized dashboard. Simultaneously, the ILMU GLM-5.1 is embedded as a secondary actor that offers the critical

reasoning service, which is needed to convert these inputs to actionable intelligence. One of the technical highlights of this architecture is the "Generate Strategic Forecasts" use case, which uses a multi-step prompting approach to combine structured sales data with real-life factors such as weather and local events. The system addresses the trust gap among non-technical users by providing a special interface called Audit AI Reasoning Logs, such that every recommendation made by AI comes with an explanation that can be fully understood by humans. This design eventually transforms the SME owner into a more proactive, data-driven curator instead of a manual and reactive planner.

6. Features included (Scope Definers)

The SmartStock MVP has a narrow and clear scope to deliver a successful product in the constrained timeframe of the hackathon, and yet has good AI-based decision intelligence capabilities. The chosen features are to overcome the most burning issues of SMEs in terms of inventory management, especially the absence of data-based insights and decision support systems.

Firstly, the system allows not only CSV upload but also manual entry of data that can help in supplying structured sets of data, including historical sales records, inventory quantities, and product data. This two-input solution is flexible and convenient, allowing SMEs with different degrees of digital maturity to use it, such as those that already use spreadsheets and those that use manual record-keeping. Moreover, the users can also add contextual and unstructured data (e.g., forecasted weather, local events, or promotions), which enables the system to capture the real-world factors that affect demand. The platform is also equipped with an inventory dashboard, a centralized interface on which business performance is monitored. This dashboard provides an understandable and visualized view of the current level of stock, past sales patterns, and performance metrics of the products. The interface is easy to use and user-friendly, catering to non-technical users and allowing them to interpret data and pick up patterns quickly without the use of sophisticated analytical software.

The main value proposition of SmartStock is the AI Insights and Recommendations Panel, which is at the center of the system. Powered by the Z.This component is the AI General Language Model (GLM) that takes structured and unstructured input and provides demand forecasts and actionable inventory suggestions. These insights can help users make informed choices about stock replenishing, cutting down on surplus inventory, and preventing out-of-stock situations. The system moves the decision-making process to a proactive one by converting the raw data into useful guidance. To increase user confidence and transparency of the system, SmartStock integrates an explainability aspect of AI. Each recommendation is supported with a clear, readable explanation as to why the AI gave the output it did. This will guarantee that the users are not just taking suggestions but also comprehending the rationale behind these suggestions, which is essential to adoption and trust in AI-aided decision-making.

Lastly, any insights generated by AI are presented in a structured JSON. This allows maintaining the same format of output, makes it easier to validate responses, and allows possible future integration with other systems, dashboards, or analytics tools. This design option also guarantees that the system can be scaled and extended past the MVP.

7. Features Not Included (Scope Control)

In order to keep the project within the limits of the hackathon time frame, it is hidden that a number of advanced features are out of the scope of the current MVP. These omissions enable the team to concentrate on providing a strong and quality core system without overworking on the development. The system lacks, firstly, real-time inventory tracking with the use of IoT devices. It implies that SmartStock is not used with physical sensors and automatic stock monitoring systems to deliver real-time inventory updates. Rather, it is based on user-supplied data inputs, which are adequate to illustrate the essence of the decision intelligence capabilities of the platform.

Secondly, this version does not support multi-user collaboration. The MVP has a single-user interaction design and thus lacks features like role-based access control, shared workspaces, and concurrent multi-user sessions. This is a simplification that allows the complexities of systems to be minimized while retaining the core functionality. Also, the system lacks the advanced forecasting methods using the classic machine learning models like ARIMA, LSTM, and hybrid predictive systems. The answer is solely dependent on the Z.General Language Model (GLM) as the key reasoning engine, as required by hackathons. This makes sure that the AI aspect is the focal point of decision-making within the system.

Finally, the MVP does not cover full integration with enterprise-level systems like ERP systems (e.g., SAP, Oracle). Although these integrations would make the system more applicable in the real world, they cannot be included in a one-week development cycle. Nevertheless, these attributes are seen as possibilities to improve in the next versions, and scalability and enterprise adoption can be developed.

8. Assumptions & Constraints

I. LLM Cost Constraint & Latency Constraints

There have been significant efforts to optimize the system for being cost-efficient and efficient on API cost in relation to usage of the Z.AI GLM API.

- **Context Token Limit (Safeguard):** The backend uses a Context Token Limit of 1500 tokens. This is a safeguard from having the system incur too much cost from the API and keeping the inference time less than 5 seconds.
- **Data Chunking Approach:** The system uses a CSV Data chunk approach of 7 days to control the LLM costs and keep latencies low while being below the 1500 token constraint. Rather than passing months of raw data, the system preprocesses and chunks the structured sales data into optimized windows of 7 days before appending the data to the prompt.
- Saving frequently asked questions to avoid repeated calls to the API.

II. Technical Constraints

- **Backend Framework:** FastAPI (to handle API efficiently and integrate with GLM)
- **Frontend Framework:** React (responsive and user-friendly UI)
- **Database:** PostgreSQL (to store data in a structured and be dependable)
- **AI Requirement:** Compulsory use of Z.AI GLM as a reasoning engine.

The architecture will be light and modular, as well as capable of quick development of an MVP within a limited time.

III. Performance Constraints

- **AI Response Latency:** Around 2-5 seconds per request.
- **Input size limitation:** This will limit the size of data input to make it efficient and will not allow it to exceed the token limit.
- **System Scope:** Single-user session optimization at the MVP stage.

These constraints guarantee a balance of performance, usability, and technical feasibility.

IV. User Input Constraint

- The system solely depends on the data provided by the users, and thus, the quality of the outputs is directly proportional to the accuracy and completeness of the inputs.
- The AI-generated recommendations are supposed to assist, rather than to substitute, human decision-making.
- The users have to check and authenticate AI results before acting.

9. Risks & Questions Throughout Development

Risk Category	Potential Risk	Mitigation Strategy
Data Integrity	Users submitting conflicting or inaccurate sales data affecting AI insights.	Implement strict input validation controls and discrepancy flagging during CSV ingestion.
Model Reliability	Z.AI GLM produces "hallucinated" strategies not supported by historical facts.	Enforce structured JSON output validation and multi-step prompt engineering.
Contextual Ambiguity	AI misinterpreting vague user notes like "busy weekend" without specific metrics.	Utilize standardized prompt templates and clarification loops for uncertain inputs.
Scalability	Performance degradation when processing large datasets or complex SME portfolios.	Apply advanced data summarization and database indexing to maintain low-latency inference.